



Evaluation of PHASES Score for Predicting Rupture of Intracranial Aneurysms: Significance of Aneurysm Size

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■ **BACKGROUND:** Recent data have identified that certain risk factors for rupture differ between small and larger intracranial aneurysms (IAs). Such differing risk factors make up 5 out of the 6 predictor variables used in the PHASES score, which raises the question on whether IA size has a significant effect on the score's performance.

■ **METHODS:** Patients who were diagnosed with an IA incidentally or due to a subarachnoid hemorrhage between 2015 and 2023 were selected for potential inclusion. The median IA size of the cohort was chosen as the cutoff point to categorize small and large (6 mm). The PHASES score was calculated for all patients, and a receiver operating characteristic curve analysis was performed to evaluate the classification accuracy of PHASES in predicting rupture for small and large IAs.

■ **RESULTS:** A total of 677 IAs were included. Among the IAs, 400 (58.9%) presented as UIAs and 279 (41.0%) as subarachnoid hemorrhage. The average PHASES score was 2.9 and 6.5 for small ($n = 322$) and large ($n = 355$) IAs, respectively. The PHASES score performed significantly lower for predicting rupture in smaller IAs (area under the curve: 0.634) compared with the larger (area under the curve: 0.741) ($P = 0.00083$).

■ **CONCLUSIONS:** PHASES was shown to underperform on small IAs. The decision to treat small unruptured IAs

remains highly controversial, and the development of a new score to estimate the annual rupture rate while accounting for IA morphology is of great need. Our findings can help encourage future researchers to develop such a score.

INTRODUCTION

The decision to treat an unruptured intracranial aneurysm (IA) can often be controversial. The natural history has been well studied after the release of 2 major largescale prospective observational studies: The International Study of Unruptured Intracranial Aneurysms (ISUIA)^{1,2} published in 1998 and the Unruptured Cerebral Aneurysm Study of Japan (UCAS)³ in 2012. Despite having a stronger knowledge regarding the natural history, or in other words, the annual rupture rate for various types of IAs, there still remained no scoring system until the mid-2010s, which could allow for a less opinion-based treatment decision making process for unruptured IAs. In 2014, PHASES became the first major scoring system that sought to alleviate this issue.⁴ PHASES is, in essence, a meta-analysis that obtained the individual patient data of the 6 prospective unruptured IA studies within the literature at its time to identify independent predictor variables for rupture to develop a score that could estimate the 5-year rupture rate. The score consists of 6 predictor variables assigned with various weights (Table 1) and to

Key words

- Intracranial aneurysms
- PHASES
- Risk factors
- Rupture
- Size

Abbreviations and Acronyms

CTA: Computed tomography angiography

IA: Intracranial aneurysm

ISUIA: International Study of Unruptured Intracranial Aneurysms

ROC: Receiver operating characteristic

SAH: Subarachnoid hemorrhage

UCAS: Unruptured Cerebral Aneurysm Study of Japan

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Citation: World Neurosurg. (2024) 184:e178-e184.

<https://doi.org/10.1016/j.wneu.2024.01.077>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

Available online: www.sciencedirect.com

1878-8750/\$ - see front matter © 2024 Published by Elsevier Inc.

date remains as the largest and highest quality scoring system that estimates aneurysm rupture. However, with time, PHASES has received significant criticism over its design and limitations. According to the literature since the score's release, one of the main issues is that PHASES has consistently been shown to fail to provide clear recommendations for the management of most aneurysms that eventually rupture and to agree with the consensus view of multidisciplinary cerebrovascular teams.⁵⁻¹¹ The second more recent concern is the cumulating data suggesting the aneurysm morphology is outperforming the standard patient demographic variables when used to predict rupture. The study by Varble et al¹² revealed that small (<5 mm) and large (≥5 mm) IAs do not share the same clinical predictors (e.g., age, previous subarachnoid hemorrhage [SAH], hypertension, IA location); however, this discrepancy is not seen for IA morphology variables. Given these findings and that 5 out of the 6 predictors in PHASES are clinical predictors, it would be of great interest to assess the performance of PHASES on a stratified cohort of small and large IAs.

Table 1. Predictors, Criteria, and Points Composing PHASES Score to Estimate 5-Year Aneurysm Rupture Rate

Criteria	Points
Population	
North American, European (other than Finnish)	0
Japanese	3
Finnish	5
Hypertension	
No	0
Yes	1
Age	
<70 years	0
≥70 years	1
Size of aneurysm	
<7.0 mm	0
7.0–9.9 mm	3
10.0–19.9 mm	6
≥20.0 mm	10
Earlier SAH from another aneurysm	
No	0
Yes	1
Site of aneurysm	
ICA	0
MCA	2
ACA/PcoA/posterior circulation	4
SAH, subarachnoid hemorrhage; ICA, internal carotid artery; MCA, middle cerebral artery; ACA, anterior communicating artery; PcoA, posterior communicating artery.	

It is worth mentioning the fact that given PHASES utilized all prospective datasets available, and that no other prospective natural history trial had commenced since, no external validation of the score has ever been conducted. Thus in the past 5 years, there has been a growing interest in validating the PHASES score in an alternative manner.¹³⁻¹⁸ Limited to only retrospective chart reviews, researchers have, instead of validating the 5-year estimation of PHASES, rather validated the score's performance when used as a dichotomous classifier for predicting rupture using the receiver operating characteristic (ROC) curve, which is a common approach to alternatively validate scores.¹³⁻¹⁷ Such methods do have their limitations and should not be used to fully gauge the reliability of any scoring system; however, in the case for PHASES, it has become quickly normalized in the literature.¹³⁻¹⁸ As mentioned, these studies used retrospectively collected datasets. As a result, the ruptured group did not comprise IAs that presented unruptured and ruptured during follow-up but rather represented IAs that presented SAH. The control group, however, remains the same as they are those that presented incidentally. The purpose of the present study is to assess the performance of PHASES in the same manner but on a cohort of IAs stratified by size. If a significance lower difference in performance is seen in either subgroup, the findings may suggest that predictor variable modifications, or the development of a new scoring system built on datasets of solely small or large IAs, may be of need.

METHODS

Data Extraction and Exclusion Criteria

This study was performed at the University Hospital of the Medical University of Gdansk, Poland. All patients diagnosed with an IA on diagnostic imaging between January 1, 2015, and January 1, 2023, were extracted retrospectively and evaluated for potential inclusion. Exclusion criteria included (1) patients with one or more fusiform, mycotic, or dissection IAs; (2) patients with other vascular diseases such as arteriovenous malformation or moyamoya disease, among others; (3) insufficient diagnostic imaging quality to evaluate IA size; and (4) patients with one or more IAs previously treated at other hospitals.

Data Collection

The following patient characteristics are to be retrieved from hospital records: age, sex, hypertension (blood pressure >140 mm Hg systolic or on hypertensive medications), history of SAH, and symptoms at admission. If a patient had multiple aneurysms, the largest of these aneurysms, along with its location and maximum diameter (the maximum distance of any 2 points on the aneurysm), the imaging modality used for measuring the aneurysm (digital subtraction angiography, magnetic resonance angiography [MRA], or computed tomography angiography [CTA]), the outcome (treatment or conservative management via scheduled follow-up), whether treatment was initiated, and the type of treatment (surgical clipping or endovascular) were collected. IAs were subgrouped into unruptured and ruptured. Unruptured IAs were those that presented incidentally due to imaging for another reason or due to symptoms caused by IA mass effect. The ruptured group were those that presented as an SAH. The diagnosis of SAH at admission was confirmed by computed tomography (CT) based

on a pattern of hemorrhage or a lumbar puncture. The ruptured IA was confirmed by a definitive hemorrhage pattern or a craniotomy treatment.

Statistical Analysis

The total cohort of aneurysms was stratified by size. To mitigate bias, the median size of the aneurysm diameter (mm) was utilized as the cutoff point to classify small and large aneurysms. The cutoff point was later found to be 6 mm, which allowed a close to even split (322:355). In this way, the receiver operating characteristic curve analyses would have near to even statistical power for both small and large aneurysms in terms of their sample sizes. All collected continuous variables such as patient age and aneurysm size are presented as means $\pm$ standard deviations, while categorical variables are presented as percentages. T-test and Mann-Whitney U test were used for continuous variables, and a chi-square test was used for categorical variables. ROC curves are empirical curves in the sensitivity and specificity space¹⁹. An area under the curve (AUC) of 0.5 suggests no discrimination, while an AUC of 1 suggests perfect discrimination. Statistical difference between the small and large aneurysm ROC curves was tested with de Long's method.²⁰ 95% Confidence intervals (CIs) of sensitivity and specificity were computed with bootstrap resampling and averaging methods.²¹ All statistical tests are 2 sided, and statistical significance is defined as $P < 0.05$. Finally, to the best of our knowledge, at present 6 studies have evaluated the performance of PHASES for predicting rupture. These studies were pooled along with the present study in a forest plot using random effects to obtain the overall AUC from

their ROC curve analyses. All statistical analyses and plots generating were performed with R studio (version 1.4.1717), and the following packages were utilized: readxl, pROC, ggplot2, and metafor.

RESULTS

A total of 677 patients met the inclusion criteria and were involved in the study. Among these patients, 492 (72.67%) were female and the mean age was 58.39 ± 14.1 years. Among the 677 aneurysms, 58.9 were unruptured and 41.1% were ruptured. Patient history of a prior SAH was low (2.7%), and 53.5% were classified as having hypertension. Regarding IAs, the average size was 7.0 ± 4.5 mm and 16.0% had multiple aneurysms (the distribution of IA size is shown in Figure 1A). The most common IA location was at the internal carotid artery (ICA) (30.6%), following the middle cerebral artery (29.1%) and the anterior communicating artery (27.2%). The most common image performed, which was later used for measuring the aneurysm, was CTA (51.3%). Aneurysms were treated in 63.8% of cases with surgical clipping being the main approach utilized (51.6%). The average PHASES score for the entire cohort was 4.8 ± 3.2 points (Figure 1B shows the PHASES score distribution stratified by UIA, SAH, and IA size). All variables are stratified by aneurysm size in Table 2.

The ROC curve analysis reported a significantly lower classification accuracy by the PHASES score for small IAs (AUC: 0.634) compared with larger ones (AUC: 0.741) ($P < 0.001$). The AUC for the total cohort was 0.62 (95% CI 0.560–0.680) (Figure 2.). Regarding the meta-analysis, the characteristics of the 6 studies yielding a total of 3788 additional IAs are reported in Table 3. The

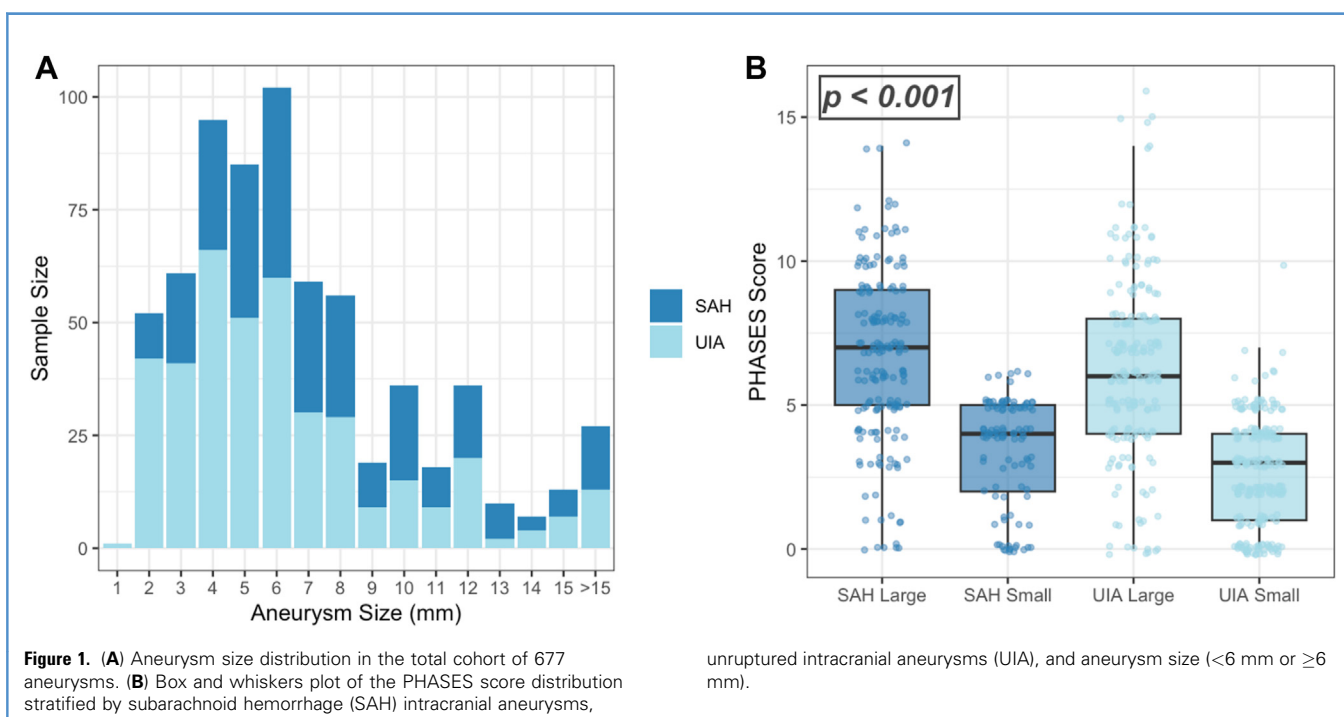
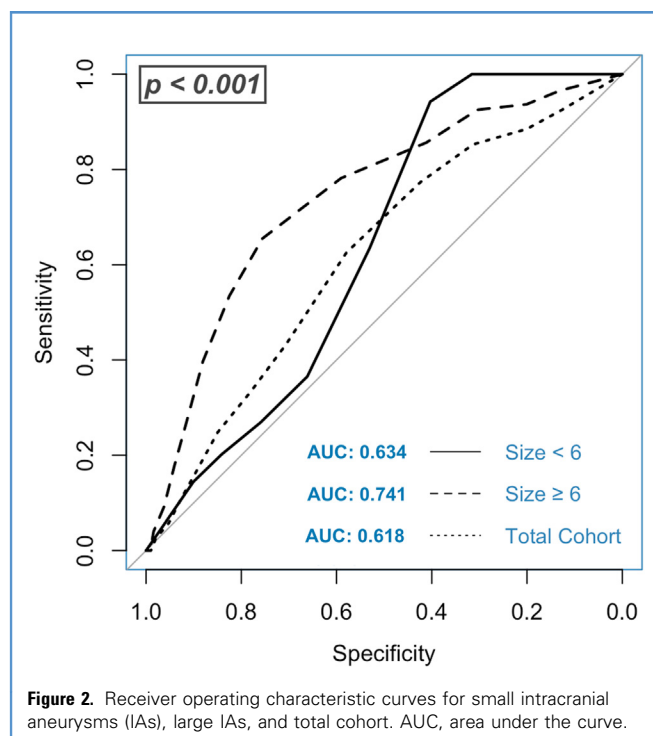


Table 2. Characteristics of 677 Patients Included in Study

Variables	Total (n = 677)	<6 mm (n = 322)	≥6 mm (n = 355)	P Value
Presentation				
Unruptured	399 (58.94%)	218 (67.70%)	181 (50.99%)	<0.01
Ruptured	278 (41.06%)	104 (32.30%)	174 (49.01%)	—
Clinical variables				
Age (years)	58.39 ± 14.05	57.74 ± 13.94	58.97 ± 14.14	0.17
Female	492 (72.67%)	237 (73.60%)	255 (71.83%)	0.87
Prior SAH	18 (2.66%)	9 (2.80%)	9 (2.54%)	0.73
Hypertension	362 (53.47%)	183 (56.83%)	179 (50.43%)	0.06
Multiple aneurysms	108 (15.95%)	39 (12.11%)	69 (19.44%)	<0.01
Location				
ACOM	184 (27.18%)	90 (27.95%)	94 (26.48%)	0.149
ACA	29 (4.28%)	15 (4.66%)	14 (3.94%)	—
MCA	197 (29.10%)	95 (29.50%)	102 (28.73%)	—
ICA	207 (30.58%)	96 (29.81%)	111 (31.27%)	—
PCOM	10 (1.47%)	7 (2.17%)	3 (0.85%)	—
PCA	7 (1.03%)	0	7 (1.97%)	—
VA	8 (1.18%)	4 (1.24%)	4 (1.13%)	—
BA	28 (4.14%)	10 (3.11%)	18 (5.07%)	—
CA	7 (1.03%)	5 (1.55%)	2 (0.56%)	—
Imaging modality				
DSA	296 (43.72%)	137 (42.55%)	159 (44.79%)	0.77
CTA	366 (54.06%)	170 (52.80%)	184 (51.83%)	-
MRA	15 (2.22%)	15 (4.66%)	12 (3.38%)	-
Aneurysm management				
Surgical	349 (51.55%)	134 (41.61%)	215 (60.56%)	<0.01
Endovascular	80 (11.82%)	27 (8.39%)	53 (14.93%)	—
Observed	248 (36.63%)	161 (50.00%)	87 (24.51%)	—
PHASES				
Score	4.78 ± 3.17	2.89 ± 1.91	6.49 ± 3.12	<0.01

SAH, subarachnoid hemorrhage; ACOM, Anterior communicating artery; ACA, anterior communicating artery; MCA, middle cerebral artery; ICA, internal carotid artery; PCOM, posterior communicating artery; PCA, posterior cerebral artery; VA, vertebral artery; BA, basilar artery; CA, cerebellar artery; DSA, digital subtraction angiography; CTA, computed tomography angiography; MRA, magnetic resonance angiography.

overall sample sizes ranged from 381 to 1004 and 4 out of the 6 were from the Chinese population, whereas the remaining 2 were Swiss or Finnish. Five of the 6 were retrospective. The pooled AUCs in the random effects forest plot reported a value of 0.650 (95% CI 0.630–0.670) (Figure 3). Interestingly, all studies had reported similar AUC results, suggesting that the



overriding AUC obtained by the plot is highly reliable. In addition, study heterogeneity was not significant ($P = 0.556$).

DISCUSSION

The present study identified that the PHASES score significantly underperforms on smaller (<6 mm) aneurysms when predicting rupture. This finding is indeed of great interest to cerebrovascular specialist given that small aneurysms have remain poorly understood regarding their natural history compared to their larger counterparts. This is due to the conflicting results reported by the UCAS and ISUIA trials. As mentioned, both trials have been the major sources used to base decision-making for UIAs. Both prospectively assessed the annual rupture rates of UIAs of various sizes and found similar results. However, for small aneurysms the 2 reported conflicting results. ISUIA identified a risk of hemorrhage for aneurysms <7 mm to be 0% over 5 years for group 1 patients (single unruptured lesion) in the anterior circulation and 1.5% for group 2 patients (prior subarachnoid hemorrhage [SAH] from another aneurysm). For posterior circulation lesions, the rates were 2.5% over 5 years for group 1 patients and 3.4% for group 2 patients.² UCAS had a much higher percent of smaller aneurysms with an overall annual risk of hemorrhage of 0.36% for lesions 3–4 mm.³ The conflicting results between the two trials has made the decision-making process for smaller unruptured IAs difficult and controversial. As a result, reevaluating common and establishing new predictor variables for rupture have been a growing interest in the literature especially for small aneurysms.^{12,22–35} On a side note, it is shown in the literature that the

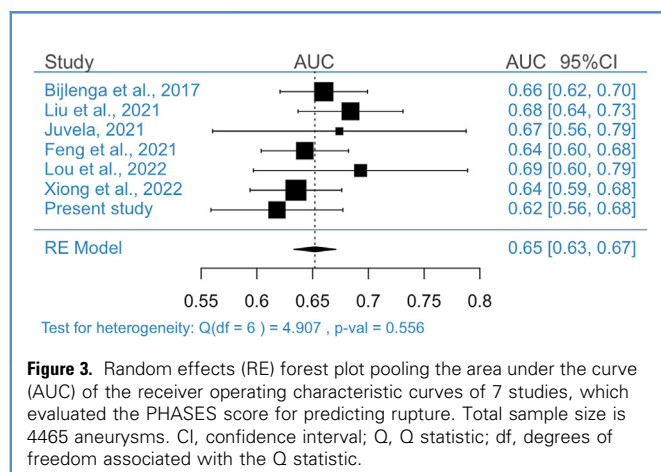


Figure 3. Random effects (RE) forest plot pooling the area under the curve (AUC) of the receiver operating characteristic curves of 7 studies, which evaluated the PHASES score for predicting rupture. Total sample size is 4465 aneurysms. CI, confidence interval; Q, Q statistic; df, degrees of freedom associated with the Q statistic.

cutoff point in size to define smaller and larger aneurysms is highly inconsistent and can lead to confusion to readers. For example, we see that “small” has been defined as <5mm in ISUIA 1998, <6mm in the present study, and <7 mm in UCAS. This inconsistency should not stumble readers but rather suggest that because the controversy of treating an aneurysm also depends on several other factors, we encourage readers when coming across the word “smaller” or larger” when describing aneurysms, to treat the terminologies as slightly ambiguous in this article rather a cutoff-point unless specified. We have known since Varble et al’s¹² study in 2017, that small and large IAs do not share the same clinical predictors (such as age, previous subarachnoid hemorrhage (SAH), hypertension, and IA location).³⁶ Given that IA morphology variables (such as aspect ratio, size ratio, etc.) on the other hand lacked such discrepancies seen in the clinical ones raises much more interest in the modification of the PHASES to include such aneurysm parameter measurements in order to mitigate performance bias towards aneurysms of various sizes.

Past Modifications to PHASES and New Scores

Modifications in the form of adding new predictor variable to PHASES has already been attempted.¹⁸ In 2021, Juvela addressed the controversy on why smoking was not included in PHASES which has been long regarded as an important factor for IA rupture given the findings from the UCAS and ISUIA trails. As a result, Juvela investigated whether the inclusion of the variable, smoking, would enhance the classification accuracy of PHASES when used to predict rupture. As suspected, the modified score with smoking included reported an improved AUC (0.755, 95% CI 0.657–0.853) compared to PHASES (0.674, 95% CI 0.558–0.790) when applied onto a prospective dataset of 142 Finnish patients (note: this dataset was 1 of the 6 prospective datasets used in the development of PHASES). However, there are other studies in the recent years which took an additional step in developing their own score with a focus on morphology related predictors.^{14,16,17,34} Unlike the study by Juvela, these scores were derived from retrospective datasets and thus should be interpreted with caution. We know that aneurysm morphology not only are shared predictors for both small and large aneurysms but have consistently shown to outperform clinical predictors in the literature.^{12,22–35} Thus, these pilot studies which have developed their scores that allocate significant weight to morphology, are also reporting substantially improved performance compared with PHASES. Liu et al in 2021 was the first of its class to develop and validate its model, which reported a significantly improved AUC (0.837 [95% CI 0.780–0.894]) compared with PHASES (0.657, [95% CI 0.585–0.729]) on their dataset of 658 aneurysms ($P < 0.001$).¹⁴ The same and following year yielded other similar scores, which focused on solely small aneurysms^{16,34} or utilizing machine learning,¹⁷ all which involved mainly morphologic predictors. Due to their study design being retrospective in nature, it is needed to validate their results with prospective data before clinical implementation.

Considerations for Future Score Developers

According to trends in the literature, it is clear the aneurysm morphology is becoming a focus of interest among researchers

Table 3. Literature Review of All 7 Studies Evaluating PHASES Score for Predicting Intracranial Aneurysm (IA) Rupture

Study	Country	Design	N	Main Objectives	AUC
Bijlenga et al., 2017 ¹³	Switzerland	Retrospective	841	Evaluate PHASES ability to match specialists’ decisions, classify patients with ruptured or unruptured IA, and discriminate low–rupture risk patients	0.660
Liu et al., 2021 ¹⁴	China	Retrospective	719	Development and validation of a novel nomogram for aiding aneurysm rupture risk stratification	0.684
Juvela, 2021 ¹⁸	Finland	Prospective	142	Enhancing the PHASES score by including smoking as a predictor variable (Note: same dataset used in PHASES)	0.674
Feng et al., 2021 ¹⁵	China	Retrospective	701	Evaluating PHASES predictive performance on multiple IAs	0.643
Lou et al., 2022 ¹⁶	China	Retrospective	381	Development of a novel nomogram for aiding small IA–rupture risk stratification	0.693
Xiong et al., 2022 ¹⁷	China	Retrospective	1004	Development of a novel machine learning model to predict rupture of small IAs	0.635
Present study	Poland	Retrospective	677	Evaluating PHASES predictive performance on small and large IAs	0.618

N, total cohort sample size; AUC, area under the curve.

seeking to develop enhanced scoring systems for rupture prediction. As a word of caution, using retrospective data to build such systems where the ruptured group were obtained as IAs that presented SAH, and the unruptured group being IAs which presented incidentally should be strongly discouraged. This approach in the past decade has become quite normalized which has allowed for rapid analyses and the growth of inventing new morphological parameters showing strong associations with rupture.^{12,22–35} However, we now know as of earlier this year that such approaches should be interpreted with caution.³⁷ Fujimura et al³⁷ showed that before rupture, the IA morphology is significantly different compared with post rupture. Indeed, this makes intuitive sense regarding to what degree of change occurs had never been sufficiently assessed until the release of Fujimura et al's³⁷ findings. In brief, the main changes after rupture are an alteration into an elongated and irregular geometry, accompanied by an increase in aneurysm volume. Such factors would affect most of the various morphology related predictors for rupture derived from retrospective datasets. Thus the current question is to what degree is the exceedingly high statistical significance we see in these retrospective studies due to the actual association of these measurements themselves with rupture, or rather, is the significance attributed by the study design bias? This question will require thorough investigation; however, clinicians who choose to use scores should be advised to only utilize the adjuncts that have been derived from large-scaled, prospectively collected unruptured IAs.

Limitation

The present study contains several limitations. First, only a Polish population (classified as North American, European [other than Finnish] in PHASES) was involved, which warrants the need for future studies to validate our findings with a Japanese and Finnish population. However, the meta-analysis does provide some insight that our findings are likely generalizable given that the total cohort reported similar AUC results compared with studies involving Finnish and other non-Finnish European populations. Finally, aneurysm measurement was performed using a broad range of imaging modalities. It is likely that our measurements for the maximum aneurysm diameter may be affected given the variations

in image resolution among CTA, DSA, and MRA. Furthermore, IA measurement was performed by several neurosurgeons spanning a wide range in years of clinical experience (3–20 years). However, it has been shown that a neurosurgeon's years of experience does have a significant effect on the precision on aneurysm measurements on diagnostic imaging.³⁸

CONCLUSION

The present study is the first to investigate and report that aneurysm size has a significant effect on PHASES score performance for predicting rupture. At present, the treatment decision making for small aneurysms has several pitfalls, making the process difficult and often controversial.³⁹ The findings from this study suggest that in addition to the discrepancies seen in the natural history for small unruptured IAs from the UCAS and ISUIA trials, PHASES is likely not aiding the controversial treatment decision making process for IAs at the smaller end of the spectrum as initially hoped. However, we strongly encourage future prospective studies to validate and build off of our findings.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Samuel D. Pettersson: Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft. **Paulina Skrzypkowska:** Data curation, Writing – original draft, Software, Project administration. **Krzysztof Pietrzak:** Data curation, Software. **Aleksander Och:** Data curation, Software. **Kamil Siedlecki:** Formal analysis, Supervision, Writing – original draft. **Aleksandra Czaplak-Iskrzycka:** Data curation, Software, Supervision. **Tomasz Klepinowski:** Formal analysis, Supervision, Writing – review & editing. **Thomas Fodor:** Formal analysis, Writing – original draft. **Jean Filo:** Formal analysis, Writing – original draft. **Jaroslav Meyer-Szary:** Formal analysis, Software, Supervision, Writing – review & editing. **Justyna Fercho:** Software, Supervision, Writing – review & editing. **Fanny Sunesson:** Supervision, Writing – review & editing. **Hanna K.L. Olofsson:** Supervision, Writing – review & editing. **Shan Ali:** Supervision, Writing – review & editing. **Tomasz Szmuda:** Supervision, Writing – review & editing. **Grzegorz Miekisiak:** Project administration, Supervision, Writing – original draft, Writing – review & editing.

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Conflict of interest statement: The authors declare that the article content was composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Received 15 October 2023; accepted 13 January 2024

Citation: *World Neurosurg.* (2024) 184:e178-e184.

<https://doi.org/10.1016/j.wneu.2024.01.077>

Journal homepage: www.journals.elsevier.com/world-neurosurgery

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